Discrete Data Interoperability from Mobile Medical Agencies (Fire & Ambulance Services)

An Undersung Story of Information Sharing Success that Needs to be Told to Avoid Reinventing the Wheel…Again

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AUTHOR'S NOTES:

1. **This white paper (FILE CODE: CMS-0042-NC) addresses RFI: "Health Technology Ecosystem" issued 16 May 2025 by the Centers for Medicare & Medicaid Services, Assistant Secretary for Technology Policy/Office of the National Coordinator for Health Information Technology, Dept. of Health & Human Services. Specifically, we address items PC-8, PC-10, PC-11, PC-12, PC-13, PR-12, PR-13, and PR-14.**

2. **In this response, we may alternately refer to ambulance-based practices as "Mobile Medicine," "EMS," "Emergency Medical Services," and "medical transportation" (and variants thereof). We intend to include all variations of Mobile Medical services, including, but not limited to, fire services, public and private ambulance services, emergency and non-emergency medical transport services, critical care transport (both ground and air), and community paramedicine / mobile integrated health.**

3. **Portions of this RFI response have been previously published by Jonathon Feit (the author) in the *Journal of Emergency Medical Services (*online at www.jems.com*)* and *Becker's Hospital Review* (http://www.beckershospitalreview.com).**

PC-8. In your experience, what health data is readily available and valuable to patients or their caregivers or both?

On January 22, 2025, the *Los Angeles Times* published "Many residents with disabilities can't flee fires on their own. Could a database help?" The reporter, Rachel Ellis, cited Los Angeles County Fire Chief Anthony Marrone as saying, during a meeting of the Board of Supervisors, that "the ongoing wildfires underscored the need for a 'database to track people who have mobility challenges or health challenges [because] trying to do it during an emergency is very, very difficult."

My father, Harold, is a diabetic transplant patient living in Los Angeles. His medications make him unable to drive, so my mother—an angel who previously cared for all four of my grandparents *while* co-running the family business *and* raising my sisters and me—takes the wheel, literally. During the 2025 Palisade Fire, I flew down from the Bay Area to evacuate them, and for the first time, Mom seemed to realize that if anything happened to *her*, we could lose them both. Why? She knows Dad's meds. If she were hurt by a fire, a car crash, or a medical crisis, Chief Marrone's warning could prove true. Many families have a Brenda who safeguards details about who needs what special care. Of course, many other families are not so fortunate: for them, a database to fill in the knowledge gaps would be enormously helpful.

The trouble comes from the fact that both the City and County of Los Angeles have *several* health information exchanges, including a native data sharing resource called the Los Angeles Network for Enhanced Services (LANES), which is funded in part by L.A. Care Health Plan, the county's largest health plan—which includes Medicaid plans used by many elderly and disabled patients, and those patients disproportionately rely on emergency services everyday (not just during disasters).

In addition, both the city and county are crisscrossed by Manifest Medex, California's largest health information organization; a federally recognized network connecting hospitals that use the Epic electronic health record system (Epic QHIN); and still another separate network that links Veterans Administration hospitals (Oracle Health). These leverage both the federal Trusted Exchange Framework and Common Agreement (TEFCA) and the California Data Exchange Framework ("DXF"), the Golden State's TEFCA analogue that was set in motion by 2021 by AB133 and in the process of (hopefully) being bolstered via SB660, to include additional governance and participation mandates for non-facility-based care providers. If those are not enough, according to Nichole Aston, Branch Chief in the California Office of Traffic Safety (OTS), a *separate* "health data exchange" project is underway in Los Angeles County, using separate funds provided by OTS.

Following the Palisade and Eaton fires, a field deputy in Supervisor Lindsey Horvath's office explained that on April 1, 2025, the Los Angeles County Board of Supervisors followed the Chief's statement by advancing a "motion, authored by [two Supervisors, that] calls for a report back from the County's Aging and Disabilities Department on how a registry could be implemented at the County level. Their report back is due in August." The motion is called, "Reviewing Best Practices

and Exploring the Creation of a Comprehensive Emergency Registry System for Older Adults and Persons with Disabilities in Los Angeles County." The Board of Supervisors' willingness to snap to achieve Chief Marrone's goal of a database of elderly and disabled patients is impressive—but ultimately misguided. They say "the road to hell is paved with good intentions": As an emergency medical services health data interoperability expert, I am concerned that this admirable goal could inadvertently *raise* the risk faced by patients like my father. Why? Because medicine wants patients to live a comfortable and active life. We *want* families from L.A. to be able to see the state and the world, confident that their health data can go with them and be accessed if the worst happens.

Given the glut of existing infrastructure, the trouble with L.A. County's approach is not the *idea* of creating a database of people who need extra help. It's that those technical systems already exist! The California DxF functions as a technical backbone for statewide interoperability, much like the SHIN-NY in New York and THSA in Texas (but it is distinct from true statewide HIEs such as MIHIN in Michigan, because the "pipes" are common and connected, but the individuals HIOs/HIEs provide local and/or differentiated services. Why does neither the L.A. City nor County Fire Department use the tools that are already available to them, despite obvious utility? Why build something new and potentially disconnected? The Chief's wishful statement suggests that he is unfamiliar about the investments that were already made and the data that were already captured (and are sharable). Yet to date, a dozen Mobile Medical services have signed up to, or are already interoperating with, the California DXF, suggesting that the issue is less one of technology than of awareness of applicability.

Moreover, both Los Angeles County Fire and the City of Los Angeles Fire Department have been invited to participate in the California DXF, but have not elected to do so as of this writing. Participation in the California DXF which would afford the agencies and their hospital partners discrete data interoperability (for improved care and operations, potentially including an elimination of paper handoffs at the emergency department, and reduced ambulance offload times); as well as and visibility into all manner of medical complexities, from disabilities that are electricity dependent, to rare conditions such as hemophilia that (if known) can alter emergency response, to the presence of end-of-life medical orders (e.g., POLST forms). AB133 established not only the legal basis for the California DXF, but also for an interoperable POLST registry that can be used by both hospital and emergency service personnel in the field.

Deployed in 2024, the California DXF weaves together disparate health sharing networks so hospitals, clinics, ambulance and fire services, doctors' offices, nursing homes, and myriad other care providers could securely interoperate when they are needed while trusting the quality of the information. It capstones over a decade of work, with four thousand-plus care providers signing a statewide Data Sharing Agreement (DSA) managed by the Center for Data Insights & Innovation in the California Department of Health & Human Services. Its signers include organizations that don't make clinical data sharing choices lightly, such as hospitals like Kaiser Permanente, Cedars-Sinai, Providence, Adventist, CommonSpirit, USC, and UCLA.

I worked inside L.A. County Fire during the COVID-19 public health emergency, helping to keep its personnel safe. I saw their technical capabilities and gaps, their resiliency, and their dedication to saving lives and property, lifting those who have fallen and keeping people's pain at bay. My family is

part of the L.A. community (technically under the protection of the Los Angeles City Fire Department, not County Fire). But both fire departments use the same patient record software to document what was done to and for patients. Both would similarly benefit from access to patients' health context.

Therefore, the county's instinct to build a separate, standalone — and potentially noninteroperable — local registry of patients with complex care needs could prove itself disastrous if Responders arrive to help someone that needs extra assistance, yet they lack the most recent data because it's sitting *in that system over there*. Likewise if patients travel beyond the county border so their data become inaccessible to Responders in the field. (The *New York Times* has covered this quandary in depth, in articles titled "Filing Suit for Wrongful Life" and "Paramedics Share One Tip that May Save Your Life.") Chief Marrone is right that software can flag people who need special help, whether because their lives depend on electrical equipment like ventilators or chair lifts; or because they have physical, mental or behavioral disabilities and cannot escape on their own (like the father and son that Ms. Ellis reported in her article, who tragically died *together* in the Eaton Fire).

However, the key to accelerating health access and health equity is having the state's largest municipalities elect to participate in the *existing* DXF—during disasters and beyond. Doing so would also achieve a long-sought ideological goal that is fast gaining nationwide momentum: inviting Mobile Medical professionals to the table so they can access data they need, which they have requested, and to which the government says they are entitled. The DXF gives emergency service agencies a path to be informed about patients' context, needs, and wishes while care is being provided—then to share what they saw and did fast enough for the details to be used by the next link in the care chain. Instead of considering something new that risks being *disconnected* from the rest of California's healthcare ecosystem, L.A. County should build upon the DXF to avoid residents' data stopping at the county line, and visitors' critical health data "going dark" to the crews in the field.

PC-10. How is the Trusted Exchange Framework and Common Agreement (TEFCA) currently helping to advance patient access to health information in the real world?

Emergency Medical Services and Medical Transportation data are based on the National EMS Information System, or NEMSIS, data standard, but NEMSIS is not governed by the U.S. Department of Health & Human Services. Rather, it is funded by the National Highway Traffic Safety Administration (NHTSA), within the federal Department of Transportation (DOT). Even insiders to the healthcare system are unaware of how EMS works in America—for example, did you know that ambulance services are paid by the mile?—and this disjoint is expensive in terms of dollars and lives. The trend is evolving, as I describe below, but it has real public health implications.

The distinction between HHS and DOT funding may seem nitpicky, but it is nuanced and critical. For many years, interoperability among Mobile Medical services and the electronic patient care record (ePCR) systems they use; EHRs used by hospitals; and HIEs was prevented by a combination of misunderstandings and business decisions–this is essential: the technology to share data among ePCRs, EHRs, and HIEs has been deployed in California since 2015–that have only recently been remedied thanks to a combination innovative steps taken by HIEs, and by rulings from ONC/ASTP.

In 2014, the California Health Care Foundation awarded a programmatic grant to my company (Beyond Lucid Technologies ("BLT")) to help us build a "translation engine" that converts NEMSIS data to HL7 CCDs so as to ensure that they are consumable by receiving care facilities, discrete and ideally in real-time. This grant was in service of a program in Alameda County, California, that was focused on reducing recidivism for patients with chronic care, and mental / behavioral health needs. Because the patients associated with such programs often have health challenges covered not by HIPAA per se, but by 42 CFR Part 2 (including mental health challenges and substance use issues), interoperability has long plagued Mobile Medical providers because those data were intertwined. The disjoint was fundamentally corrected by the issuance of ASTP's Final Rule on February 8, 2024.

In 2017, BLT detached our "prehospital pipe" from the underlying ePCR system and began collaborating with other Mobile Medical documentation companies–even those with whom we compete in the market with respect to other product lines–to facilitate vendor-agnostic data sharing across whole states. BLT currently is the principal entry point for discrete, EHR-consumable real-time data generated from NEMSIS to EHRs and HIEs in several states and even nationally thanks to our work with one of the first QHINs.

- Michigan (Michigan Health Information Network ("MIHIN"));
- Colorado (both CORHIO and the Quality Health Network ("QHN");
- New York (Bronx RHIO, with HealtheConnections in process and HIXNY on deck)

- Massachusetts (Mass HIWay; MEDITECH via Direct for a large hospital-based EMS agency)

- California (Manifest Medex, Serving Communities HIE, and Sac Valley Med Share in process)

- Minnesota & North Dakota (CommonSpirit Health via the North Dakota Health Info. Network)

- Two QHINs (MedAllies and Health Gorilla),

- Texas (two HIE integrations approved and in-process).
    - During COVID-19, BLT connected Fire services to the Texas Immunization Registry.

- Oregon (Reliance HIE, and the Oregon POLST Registry, including PCC's statewide EDIE).

    - In 2020, the focus on interoperability during COVID-19 expanded to include end of life medical orders. BLT powers the Oregon POLST Registry, which is used by 2000+ clinicians across Oregon and more than seventy (70) Fire & EMS agencies,, including data sharing with PointClickCare's Emergency Department Information Exchange (EDIE), and EHRs like Epic. The need for interoperability and real-time access to POLST forms, no matter where the patient is, is a function of their design: They literally pertain to life and death, and as such, have no margin for error. Real-time access to POLST forms is an increasingly common goal due in part to the so-called Silver Tsunami of aging individuals who want to live independently for as long as they can, even after a serious diagnosis (precision medicine and gene therapies are increasing the odds of surviving the illness, achieving remission, and even emerging from hospice). Patients want their wishes to be followed no matter where a healthcare crisis occurs. This movement was bolstered by AB133 (2021), which established the legal framework for a POLST registry in California.

In the examples above, interoperability is achieved using federal standards–such as CCDs–and focuses on getting data *into* the HIE for identification purposes (e.g., Master Patient Index) then subsequent data return. We have first focused on getting data *into* the HIE so the HIE can establish a relationship between the field clinician and the patient, thereby enabling data to asynchronously flow back to a clinician. MIHIN's deployment announcement includes an ideal summary of this approach:

> *"...when [an ambulance] is transporting a patient to [a hospital], the service will transmit its electronic patient care record containing the patient's clinical data, including vital signs, medical history, and social determinants of health, to BLT, a company that specializes in healthcare data exchange and analytics. BLT will convert the data into a Continuity of Care Document (CCD) and securely transmit it to MiHIN. MiHIN will leverage its network to route the CCD to [the hospital], allowing them to gain a comprehensive view of the patient's condition before their arrival. Additionally, MiHIN will establish a real-time active care relationship between the patient and [the hospital], facilitating seamless care coordination…MiHIN will create a real-time active care relationship between the patient and the Mobile Medical agency, enabling it to receive*

future discharge medication reconciliation reports from participating hospitals, improving medication management for patients. Furthermore, MiHIN will securely share the ambulatory CCD with any participating members of the patient's care team who have an established relationship with the patient within the MiHIN network."

The Mobile Medical profession's success at sharing critical patient data, discretely and in real-time, has not received nearly the degree of attention as challenges that other parts of the healthcare ecosystem have faced when trying to do the same thing. Therefore, it could be disastrous for Mobile Medical services—which including Fire, Emergency Medical Services, Non-Emergency and Interfacility Transport, Critical Care, and Community Paramedicine—if HIEs become HDUs and are "anointed" as a sole statewide entity, because some states have already shown themselves to be insular, ignoring or even rejecting calls to interoperate with Mobile Medical services operating in their coverage areas that are not now sharing data with the HIE but want to do so. Mobile Medical Services might find themselves in a "separate and unequal" purgatory if HDUs become standard operating procedure for America's health data ecosystem.

In July 2023, *Healthcare Innovation* reporter David Raths wrote: "…David Horrocks, who was then president and CEO of the Maryland-based Chesapeake Regional Information System for our Patients (CRISP), and John Kansky, president and CEO of the Indiana Health Information Exchange (IHIE), proposed the idea that every state should have a state-designated and regulated health data utility with a monopoly akin to an electric company. They noted that during the pandemic, states that already have such state-level organizations in place were able to leverage them for crucial public health needs." Horrocks and Kansky may be correct that healthcare services can better share data where there is one statewide HIE. But if homogenizing means leaving Mobile Medicine out, then communities will sacrifice details about polychronic patients in need of managed care (e.g., using models like "community paramedicine"), and areas that are acutely impacted by substance use (data about prehospital overdoses are virtually absent from state prescription drug monitoring programs).

The need for Mobile Medical data is underscored by the high number of roadway deaths, an increasing number of agencies researching blood administration in the field, and non-transport "co-response" deployment models seeking to align their datasets related to EMS, Fire, Public Safety, Homelessness, Substance Use, Mental and Behavioral Health, Hospital-at-Home, Family Services, and other Social Drivers of Health (SDOH) considerations. HDUs that already consume Mobile Medical data will surely keep doing so. But where HDUs do *not* consume Mobile Medical data, they will stay avoidably shut out until an adverse clinical encounter or even a weather disaster forces a too-late reconsideration of the value of real-time access to the crews in the field. We can *avoid* needing to react and wishing we had access to critical data. If we are going to realize together the vision of a connected and proactive Mobile Medical ecosystem, then we also need to advocate that even before HIEs crystallize into HDUs, they begin ingesting discrete Mobile Medical data so as to ensure that they incorporate Mobile Medical professionals' insights into the patient's health story.

The following is a perspective provided by **Dr. Chris Colwell, MD, Chief of Emergency Medicine at Zuckerberg San Francisco General Hospital and Trauma Center:**

"I do medical-legal reviews and almost half of the cases are somehow related to communication between EMS and the emergency department. Maybe half is a little overstating but certainly plenty of them. Areas like POLST and advance directives, where we are really struggling to get accurate information, and the mechanism of injury, social determinants of health," "It would have an impact on a number of levels: When we see a patient in the emergency department, there are a lot of pieces of critical information but no more so than we are getting ready to do a disposition and if I'm sending patients home, I need to know what situation I'm sending them to. And I can say that in a large percentage of my patients—I don't know the exact percentage because it changes somewhat by day—but…we see a lot of elderly patients in the emergency department. They may have fallen. If I'm sending them back to a good-structured, well-supported environment, that's a totally different world than if I'm sending them back to one where they're going to be by themselves, and the same environment they just fell in, and are very likely to fall again. And this is a significant impact on their lives, and that's a significant percentage of the patients we're seeing in the emergency department.

In an interview, Dr. Colwell continued: *"It's not just the elderly. For patients in the emergency department that are in extremis, the information on POLST, advance directives, wills and those types of things are critical to decisions that we're making. In many situations, there has been good work on it—but when that's not communicated to us, all that good work that has been done is for naught, because without knowing it we have to go on the assumption that nothing exists. There are additional resources—we've got a social medicine team that can apply some resources we never had available before, based on information that paramedics have but we don't necessarily. Social medicine is one of the resources that we can potentially apply. Knowing what situation the patient is going back to—particularly, a lot of urban emergency departments are facing this [even] intermittently too: you see your patient one weekend and then see them two weeks later and the situation has changed. They may be able to communicate some of that but either due to medical conditions or because it's a confusing system and it's not always easy, we can't always get accurate information from patients. This is where this information system is critical. I can access inpatient records and that is certainly helpful [but out-of-hospital data obtained from Fire & EMS] is the big missing piece that we need in the emergency department."*

PC-11. How are health information exchanges (HIEs) currently helping to advance patient access to health information in the real world?

The late Carnegie Mellon business professor, Lester Lave, taught an impactful lesson that he called the "Iron Law of Government." It was simple and matter-of-fact enough to fit on a bumper sticker: "When government gets involved, everybody loses." Lave meant that, even in unified governments where everyone is pulling in the same general direction, consensus is rare and fickle. A poignant example of one is ensuring that Fire and EMS personnel, arriving at the scene of a child with special health needs in crisis, knows what those needs are prior to arrival on-scene. There is no downside to such awareness—only upside for the patient, the family, the responders, the hospital that may eventually care for the patient, and the broader society. How many decision points are so clear?

Occasionally, however, the government's involvement generates a return on investment of time and debate that is *dramatically positive* because the status quo benefits so *few* that almost anything is better than what we have now. Among these is the difficulty that patients and care providers face in gaining access to healthcare data, including and especially the risk faced by everyone—patients, their families, responders, society—that something important will be missed or delayed. Since the HITECH Act became law as part of the American Recovery and Reinvestment Act of 2009, healthcare systems have been navigating the question of who can share what with whom and when and why and how much must it cost to do so.

These questions—plus the special privacy provisions of 42 CFR Part 2, which govern the sharing and redaction of mental health- and substance use- related patient data—have been at the heart of interoperability challenges, and the Mobile Medical profession's continued dismay that they frequently don't get outcomes reported back by hospitals. Forget that a frequent problem with the lack of data return is the lack of data to return on the hospital side…nevertheless, the question of engineering and support costs has long loomed over every discussion about adapting interfaces and workflows to share among ambulance, fire or community paramedicine services; hospital systems, clinics, and nursing facilities; the ePCR system used by each field service agency; and the electronic health record and/or regional health information exchange systems (sometimes, more than one EHR and/or HIE are in place for a single healthcare system).

Such a "clear as mud" status quo often keeps responders feeling like they cannot meaningfully participate at the healthcare table. That changed in July 2023, and it has meaningful (*pun intended!*) implications for the Mobile Medical profession, including Fire, EMS, Non-Emergency/Interfacility Transport, Critical Care Transport, and Community Paramedicine services. In July 2023, the U.S. Department of Health & Human Services' Office of the Inspector General (OIG) "released a final rule establishing the statutory penalties created by the 21st Century Cures Act information blocking requirements. The final rule does not impose new information blocking provisions. OIG incorporated regulations published by ONC as the basis for enforcing information blocking penalties. If OIG finds that an individual or entity has committed information blocking, they may be subject to up to a $1 million penalty per violation."

The Information Blocking Final Rule begins to bulldoze the economic impediments that prevent information sharing between Mobile Medical agencies, and between Mobile Medical agencies and hospitals and health information exchanges. At least four technology systems in our profession's ecosystem today appear to meet the federal definition of a health information exchange (HIE) or network (HIN). According to the Code of Federal Regulations:

> *Health information network or health information exchange means an individual or entity that determines, controls, or has the discretion to administer any requirement, policy, or agreement that permits, enables, or requires the use of any technology or services for access, exchange, or use of electronic health information:*
>
> > *(1) Among more than two unaffiliated individuals or entities (other than the individual or entity to which this definition might apply) that are enabled to exchange with each other; and*
> >
> > *(2) That is for a treatment, payment, or healthcare operations purpose, as such terms are defined in 45 CFR 164.501 regardless of whether such individuals or entities are subject to the requirements of 45 CFR parts 160 and 164.*

The Information Blocking Final Rule is like a data version of the prohibition on "surprise billing": organizations must know the costs associated with interoperability upfront, so they have the prerogative to accept them (or not). This change has a financial implication for agencies in the field: If a Mobile Medical agency wants to share data with a hospital or state health information exchange, the agency's ePCR vendor may be limited in its ability to do so. While the Information Blocking Final Rule lets companies charge "fees that result in a reasonable profit margin, for accessing, exchanging, or using electronic health information" in a defined set of circumstances, such charges cannot be based on competitive factors–such as trying to set up roadblocks to other companies working in a region where one vendor has historically dominated–and they cannot "put a bubble over a region" using point-to-point solutions that allow sharing of some ePCRs but not others.

Agencies also cannot be charged twice. If a Mobile Medical agency is charged for a subscription to a regional interoperability service, then the hospital cannot also be charged for it. If the hospital is charged, then the Mobile Medical agency cannot be charged. "Double dipping" has been an issue for agencies across America, for years, although such tactics are no longer allowed. Agencies are encouraged to review the guidance presented via Cornell Law School's Legal Information Institute.

Organizations that subscribe to any software that meets the federal definition of a health information exchange or network should get to know—upfront—the costs associated with interoperability. They can no longer be imposed after-the-fact; cannot be used as an impediment to collaborating with other companies; and must be "reasonable" i.e., related to explainable costs. All of which should bode well for agencies that use ePCR systems and want to make Mobile Medical interoperability a near-term reality.